Carnegie Mellon University
Mechanical Engineering

Precise Spice

Spice Container with Integrated Measuring System

Product Design, Fall 2025

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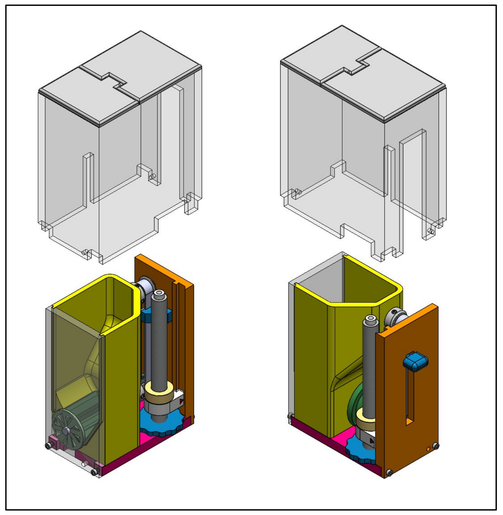
Opportunity:

Market research shows that measuring spoons often go unused in the kitchen because they require additional dishwashing and don't always fit inside spice containers. This discovery highlights a market opportunity for a premium spice container with an integrated measuring and dispensing system.

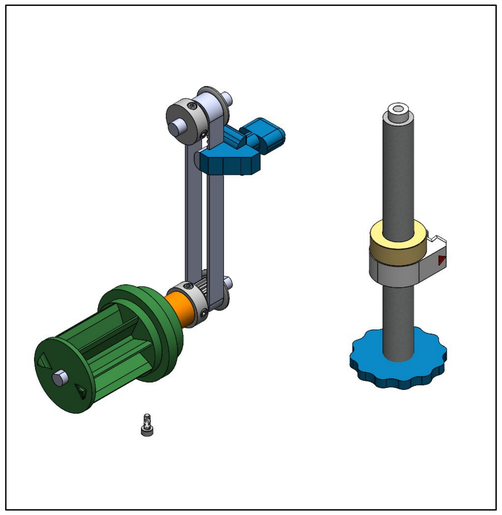
Design Solution:

The Precise Spice container lets you set and dispense your desired quantity of seasoning directly from the jar in a simple 2-step process. A lead screw lets the user set the dispense quantity with their thumb using a dial. A belt and pulley system converts user's downward force on a slider tab into the rotation of a segmented drum with ¼ teaspoon partitions, which is fed with spice by a refillable hopper. Our solution has the following qualities:

- Large capacity (7.09in^3), refillable spice storage
- Five most common measurement increments: ¼ tsp, ½ tsp, 1 tsp, ½ tbsp, and 1 tbsp
- Dispenses all spice increments via same mechanism in roughly the same amount of time
- Similar size and shape to traditional spice containers (1.8" x 3.2" x 4")

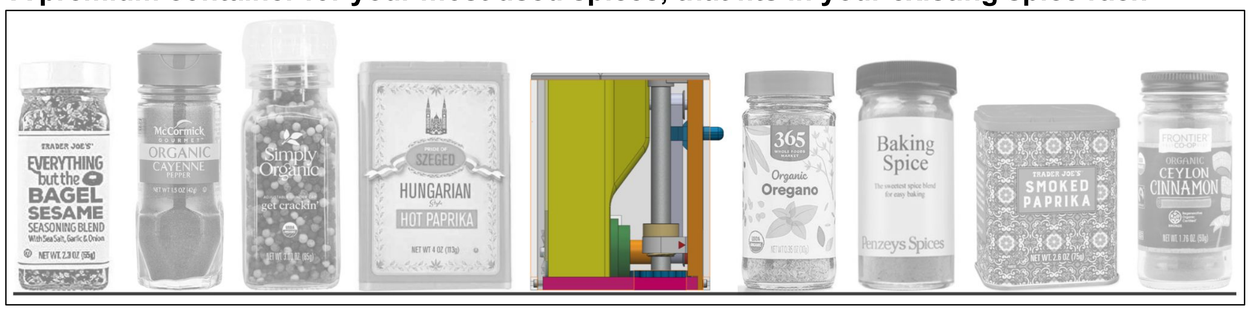


Precise Spice Final CAD



Belt & Pulley, Lead Screw Assembly

A premium container for your most used spices, that fits in your existing spice rack



Market Research:

- Measuring spoons are a hassle and therefore homecooks avoid using them
- The container needs to be able to fit in drawers and cabinets of various sizes
- Precise Spice needs to dispense the five most common measurement increments
- The spices should not leak and stay fresh for an extended period of time



Competitor Product: Select-A-Spice





Anthropological Study

Benchmarking:

Competitor Categories: innovative measuring spoons, containers with measuring dispenser, and AI spice mix makers

Main Competitor: spice containers with integrated measuring dispenser

Market Gap: existing solutions can only dispense ¼ tsp of spice at a time and are inaccurate and unintuitive

Engineering Analysis:

Belt Length Calculations:

To dispense $1\ tbsp = 1.5\ rotations$

Pulley diameter: $d_p = 0.382in$

Required linear travel: $\triangle s = 1.5\pi d_p = 1.8in$

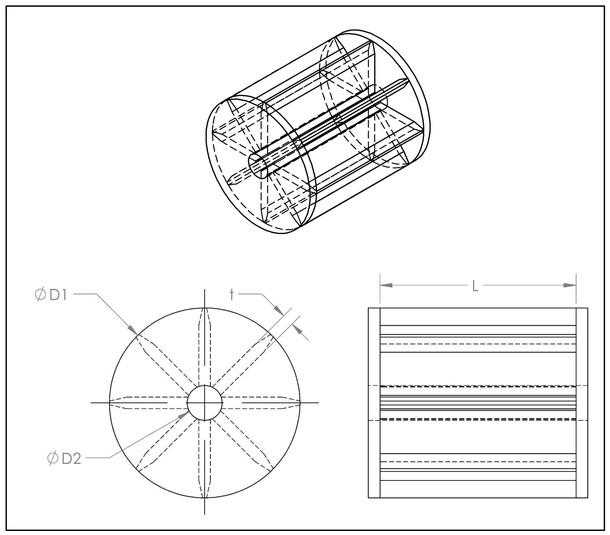
Minimum belt length: $l_b = 2\triangle s + \pi d_p = 4.8in$

Drum Volume Calculations:

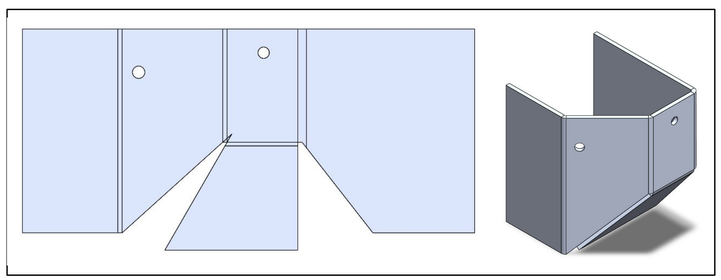
$$V_{total} = 0.25D_1^2L = 0.824668D_1$$

$$V_{hole} + V_{walls} = 0.25\pi D_2^2 L + 8tL\frac{D_1 - D_2}{2}$$
$$= 0.2625D_1 - 0.020226$$

$$V_{total} - V_{hole} - V_{walls} = 8(0.0752in^3)$$
$$\rightarrow D_1 = 1.014in$$



Drum Schematic



Sheet Metal Model of Spice Hopper

Future Work:

- Test different manufacturing techniques (e.g. sheet metal bending, injection molding, etc.)
- Improve air-tight seal
- Sprinkle and pour lid
- Decrease footprint to capacity ratio